

Construct Uniform Parity Array I & II (Leetcode 3875 & 3876) - Lecture Notes

1. Motivation: The Price of Success

"The hard days are part of the price... Become the person your goal requires." Dosto, welcome back to the channel! Aaj hum Leetcode ke do bohot hi interesting problems solve karne waale hain: **Construct Uniform Parity Array I (Easy)** and **Part II (Medium)**. Mujhe pata hai aapne shayad Part I solve kar liya hoga, par asli maza Part II mein hai. Don't worry, hum intuition ke saath deep dive karenge taaki aapko ye concepts hamesha ke liye "grok" ho jayein. Bas yehi toh game hai—humein sirf code nahi likhna, logic ko "feel" karna hai. Let's get started!

2. Understanding Part 1 (Leetcode 3875): The Freedom of Parity

Part 1 mein humein ek array $nums1$ di gayi hai aur humein $nums2$ construct karni hai. Hamara goal ye hai ki $nums2$ ke saare elements ya to **All Odd** honge ya **All Even**. Humein har element ke liye do choices di gayi hain:

1. **Keep it as is:** $nums2_i = nums1_i$
2. **Modify it:** $nums2_i = nums1_i - nums1_j$ (Jahan $j \neq i$)

The Parity Rules

Sabse pehle ye simple mathematics samajhte hain ki subtraction parity ko kaise badalta hai: | Operation | Result | Example || ----- | ----- | ----- || **Even - Even** | **Even** | $4 - 2 = 2$ || **Odd - Odd** | **Even** | $5 - 1 = 4$ || **Even - Odd** | **Odd** | $4 - 1 = 3$ || **Odd - Even** | **Odd** | $5 - 2 = 3$ |

Core Parity Flipping Observation

Dhyan se dekho table ko! Agar aapko kisi number ki parity flip karni hai (Even ko Odd banana hai ya Odd ko Even), toh aapko usme se ek **ODD** number subtract karna hoga.

- **Subtracting an ODD number flips the parity.**
- Subtracting an EVEN number keeps the parity same.

Conclusion for Part 1: Why it's always True?

Part 1 mein answer hamesha **True** hota hai. Kyun? Kyunki humein koi "strictly smaller" constraint nahi diya gaya.

- Agar array mein ek bhi Odd number hai, toh hum usse use karke baaki saare Even elements ko Odd bana sakte hain.
- Agar saare elements Even hain, toh wo already "Uniform Parity" follow kar rahe hain.
- Freedom itni zyada hai ki hum hamesha koi na koi combination dhund hi lenge. **Transition:** Lekin dosto, Part II mein "freedom" cheen li gayi hai. Ek naya constraint poora game badal deta hai.

3. Understanding Part 2 (Leetcode 3876): The "Strictly Smaller" Constraint

Part 2 mein twist ye hai: **$nums1_i - nums1_j \geq 1$** Iska simple matlab hai: Aap $nums1_j$ tabhi subtract kar sakte ho jab wo $nums1_i$ se **strictly smaller** ho.

Wait, Why the Smallest Element? (The "Grok" Moment)

Socho, agar aapko kisi number ki parity badalni hai, toh aapko ek **smaller Odd number** chahiye. Par agar aap array ke **sabse chote element (Minimum Element)** ko dekho... usse chota toh koi hai hi nahi! Iska matlab, **Minimum Element ki parity hum kabhi change nahi kar sakte**. Wo jaisa hai, waisa hi rahega. Wo hamara bottleneck hai.

Deep-Dive Analysis

Ab do hi raaste hain: **Case A: Making everything ODD** Humein saare Even numbers ko Odd banana hoga ($\$Even - Odd = Odd\$$). Iske liye humein us Even number se chota ek Odd number chahiye.

- **Insight:** Agar poore array ka **Minimum Element Odd** hai, toh hum use "weapon" ki tarah use karke baaki saare bade Even numbers ko flip kar sakte hain. Toh answer **True** ho jayega. **Case B: Making everything EVEN** Humein saare Odd numbers ko Even banana hoga ($\$Odd - Odd = Even\$$). Iske liye humein us Odd number se chota ek Odd number chahiye.
- **The Problem:** Agar array mein ek bhi Odd number hai, toh jo **sabse chota Odd number** hoga, usse chota koi Odd number milega hi nahi use flip karne ke liye!
- Iska matlab, agar array mein ek bhi Odd number hai, toh hum sabko Even **tabhi** bana sakte the jab hamare paas usse chota koi Odd element hota. Par sabse chote Odd ke liye toh ye impossible hai. **Key Insight:** Agar poore array ka **Minimum Element Even** hai, toh use hum kabhi flip karke Odd nahi bana sakte. Iska matlab hamari majboori hai ki hum poore array ko Even rakhein. Par agar array mein koi bhi ek bhi Odd number mil gaya, toh hum use Even nahi kar payenge (kyunki hamara minimum toh even hai, aur odd ko even banane ke liye smaller odd chahiye hota hai). In that case, answer **False** hoga.

4. The Unified Algorithm & Logic Flow

Ab logic chamak gaya? Let's write the steps:

- **Sabse pehle:** Array ka min_element nikaalo.
- **Check karo:** Kya min_element **Odd** hai?
- Agar **Yes**, toh Return True. (Hum is smallest odd element ko use karke baaki sabko flip kar denge).
- **Agar min_element Even hai:** Toh hum sabko Odd toh nahi bana sakte. Ab hamari last hope hai ki sabko **Even** rakhein.
- **Check karo:** Kya array mein koi bhi **Odd** element hai?
- Agar ek bhi Odd element mila $\rightarrow$ Return False. (Kyunki smallest even hai, toh is Odd ko flip karne ke liye chota Odd milega hi nahi).
- Agar koi bhi Odd nahi mila $\rightarrow$ Return True. (Saare already even hain).

Dry Run Corner

- **Example 1: 1, 4, 7**
- min_element = 1 (Odd).
- Since smallest is Odd, it can act as a base to flip 4 ($\$4-1=3\$$).
- Result: **True**.
- **Example 2: 2, 3, 8**
- min_element = 2 (Even).

- Ab humein check karna hai: Kya hum sabko Even bana sakte hain?
- Array mein '3' (Odd) baitha hai. Humein 3 ko Even banana hai, par usse chota sirf 2 (Even) hai.
- Rule yaad hai? Even subtract karne se parity change nahi hoti ($3 - 2 = 1$, still Odd).
- Isliye 3 hamesha Odd hi reh jayega aur 2 Even. Uniform parity impossible hai.
- Result: **False** .

5. Complexity Analysis

Type,Complexity,Explanation

Time Complexity, $O(N)$,Ek pass minimum element find karne ke liye aur ek pass odd element check karne ke liye ($O(2N) \approx O(N)$).

Space Complexity, $O(1)$,"Humne koi extra array ya map use nahi kiya, bas simple variables use kiye hain."

Bhaiya's Tip

Dosto, is problem ka asli hero **Smallest Element** hai. Competitive programming mein jab bhi elements ko modify karke kisi property (jaise parity) ko uniform banana ho, hamesha dekho ki smallest ya largest element kaise restrict ho rahe hain. Agar bottleneck samajh aa gaya, toh Medium problem bhi Easy ban jati hai!**Keep Coding, Keep Storytelling!**